Leakage-Safe Soccer Outcome Prediction: Chronological Fine-Tuning, Probability Scoring, and Calibration

M. Law
Final Project Submission

July 2026

Abstract

We present a reproducible prematch model for three-way soccer outcomes: home win, draw, or away win. The study normalizes 3,961 match records from StatsBomb Open Data, constructs 36 features using only information available before each match, and evaluates logistic regression, random forest, and histogram gradient boosting. The oldest 80% of matches form the training period; model and hyperparameter selection use five expanding-window folds inside that period; and the newest 20% remain untouched until final evaluation. Strongly regularized multinomial logistic regression is selected. On 793 held-out matches from October 2023 through July 2025, the final model obtains 58.89% accuracy, 0.9213 log loss, and a 0.5423 multiclass Brier score, compared with 42.24%, 1.0644, and 0.6457 for a training-prior baseline. The apparent accuracy gain is qualified by 49.06% balanced accuracy, 0.4340 macro F1, and zero hard draw predictions. Temperature scaling chosen from training-period out-of-fold predictions also worsens held-out log loss by 0.0054. The result is therefore useful as a reproducible baseline and as evidence that headline accuracy alone obscures important class and calibration failures.

1 Introduction

Soccer outcomes are intrinsically probabilistic: team strength, recent form, venue, rest, and random match events all affect the result. A useful forecast must therefore be evaluated not only by whether its most likely class is correct, but also by whether its full probability vector is informative and calibrated. Probability-sensitive scoring rules are designed for this purpose [1, 3].

This project asks: *how accurately, and with what probability calibration, can outcomes be predicted from information available before kickoff?* It extends an earlier prematch random-forest prototype and a separate submission workflow into one minimal research artifact. The main contributions are:

1. a versioned, attributed StatsBomb match table and deterministic feature builder whose state is updated only after each current row is emitted;
2. an evaluation protocol that separates the final chronological holdout from expanding-window model selection;
3. comparison and fine-tuning of three model families using log loss;
4. accuracy, balanced accuracy, macro F1, log loss, multiclass Brier score, confidence calibration, confusion, and permutation analysis; and
5. machine-readable predictions, cards, hashes, figures, and this paper.

The system is for offline research. It performs no wagering or market action and uses no audiovisual data.

2 Data Collection and Provenance

2.1 Source

The input is match metadata from StatsBomb Open Data, whose public repository provides competition-, season-, and match-level JSON for research and football analytics [5]. The inherited local snapshot contains 3,961 completed matches from 24 competitions between 24 June 1958 and 27 July 2025. The label distribution is 1,775 home wins (44.81%), 879 draws (22.19%), and 1,307 away wins (33.00%). Coverage is curated rather than population sampled and varies substantially by competition, season, era, and gender.

The final repository retains only fields needed for this study: match, competition, season, and team identifiers; display names; match date; and final scores. The source and normalized Parquet SHA-256 values begin `89690e1c2fbb` and `1d3243f77e75`, respectively. Their full 64-character values and the schema are recorded in a JSON manifest.

2.2 Labels and chronology

For home and away goals g_h and g_a, the class is

$$y_i = \begin{cases} 0 & g_h > g_a \quad \text{(home win)}, \\ 1 & g_h = g_a \quad \text{(draw)}, \\ 2 & g_h < g_a \quad \text{(away win)}. \end{cases} \tag{1}$$

Rows are sorted by date and match identifier. The first 3,168 rows (through 6 October 2023) are training data; the final 793 rows (7 October 2023 through 27 July 2025) are the test period. Unlike a random split, this setup approximates forecasting later matches from earlier history.

3 Prematch Feature Construction

Each team has state keyed by competition and season. Immediately before match i, the pipeline emits 36 numeric features in five groups:

- cumulative matches, wins, draws, losses, goals for and against, and points for both teams;
- rolling five-match points, mean goals for and against, goal difference, and window counts;
- home and away Elo ratings and their difference, with initial rating 1500 and update factor $K = 30$ [2];
- venue-specific match count, win rate, and goal difference per match; and
- capped rest days and the home-away rest difference.

For Elo, the expected home score is

$$E_h = \frac{1}{1 + 10^{(R_a - R_h)/400}}, \tag{2}$$

and the post-match update is $R'_h = R_h + K(S_h - E_h)$, with S_h equal to 1, 0.5, or 0. The key safeguard is ordering: all feature values are read first; only then are score, rolling, venue, rest, and Elo states updated with the current result. Automated tests change a current score and confirm that the current feature vector is unchanged.

State resets at each competition-season boundary. The first appearance of a team therefore has neutral Elo, zero historical counts, and a 30-day rest sentinel. Rest values are capped at 30 days to limit long off-season gaps.

4 Model Fine-Tuning

4.1 Candidates

We compare three probabilistic classifier families:

- standardized multinomial logistic regression with $C \in \{0.01, 0.1, 1, 10\}$, both with and without balanced class weights;
- a 300-tree random forest with maximum depth 6, 12, or unlimited and minimum leaf size 1, 5, or 15; and
- histogram gradient boosting with ℓ_2 regularization 0, 1, or 10.

The reference baseline is a classifier that always emits the training class prior; its hard prediction is the training majority class. All stochastic estimators use seed 42.

4.2 Selection without test-set reuse

Model selection uses five expanding-window splits over the 3,168 training rows. Each validation fold follows its corresponding training fold in time. Candidates are ranked by mean validation log loss, with accuracy only as a tie-breaker. This criterion favors useful probability distributions rather than optimizing only the most likely label. Table 1 shows the leading configurations.

The selected model is unweighted logistic regression with $C = 0.01$, indicating that strong coefficient shrinkage generalizes better across the heterogeneous training timeline than more flexible alternatives.

4.3 Temperature scaling

To assess post-hoc calibration, the selected candidate generates out-of-fold training probabilities. A temperature $T \in \{0.50, 0.55, \ldots, 2.00\}$ is chosen by minimum log loss on those predictions. Given logits z, scaling produces

$$p_k(T) = \frac{\exp(z_k/T)}{\sum_j \exp(z_j/T)}. \tag{3}$$

Table 1: Leading expanding-window validation results. Values are fold means; accuracy and log-loss standard deviations are in parentheses.

Model	Parameters	Accuracy	Log loss
Logistic	$C = 0.01$, unweighted	.500 (.076)	1.009 (.072)
Logistic	$C = 0.10$, unweighted	.491 (.086)	1.046 (.116)
Random forest	depth 6, leaf 15	.436 (.088)	1.048 (.070)
Random forest	depth 12, leaf 15	.440 (.090)	1.048 (.072)
Random forest	unlimited, leaf 15	.439 (.089)	1.048 (.073)

Table 2: Performance on 793 matches in the untouched 2023–2025 holdout.

Metric	Prior baseline	Unscaled model	Final ($T = 1.10$)
Accuracy	.4224	.5889	.5889
Balanced accuracy	.3333	.4906	.4906
Macro F1	.1980	.4340	.4340
Log loss	1.0644	.9159	.9213
Multiclass Brier	.6457	.5389	.5423
Confidence ECE	.0321	.0479	.0652

This one-parameter approach follows the temperature-scaling method studied by Guo et al. [4]. The selected value is $T = 1.10$. It is fixed before the final test probabilities are examined.

5 Evaluation Metrics

Accuracy is reported with balanced accuracy and macro F1 because class frequencies are unequal. For $K = 3$ classes, multiclass Brier score is

$$\mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K}(p_{ik} - \mathbb{1}[y_i = k])^2, \tag{4}$$

and log loss is

$$\mathrm{LL} = -\frac{1}{n}\sum_{i=1}^{n}\log p_{i,y_i}. \tag{5}$$

Lower values are better for both proper scores [1, 3]. Confidence expected calibration error (ECE) is the count-weighted absolute gap between mean maximum probability and empirical accuracy in ten equal-width bins. ECE is descriptive and bin-dependent, so it is considered alongside proper scores.

6 Results

6.1 Held-out performance

Table 2 reports the single evaluation on the chronological holdout. The final scaled model improves accuracy by 16.65 percentage points over the baseline and reduces log loss by 0.1431 and Brier score by 0.1034. However, balanced accuracy (49.06%) and macro F1 (0.4340) are materially lower than raw accuracy, revealing uneven class performance.

Temperature scaling is a negative result here. Although $T = 1.10$ minimized out-of-fold training-period log loss (1.0085 versus 1.0091 at $T = 1$), it worsened held-out log loss by 0.0054, Brier score by 0.0035, and ECE by 0.0172. This small distribution shift illustrates why calibration choices also require temporal validation and why test-time reversal would be improper.

6.2 Class behavior and calibration

Figure 1 is the central failure analysis. The model correctly classifies 290 of 335 home wins and 177 of 292 away wins, but assigns no match a draw as its maximum-probability class. Of 166 draws, 105 become home-win and 61 away-win

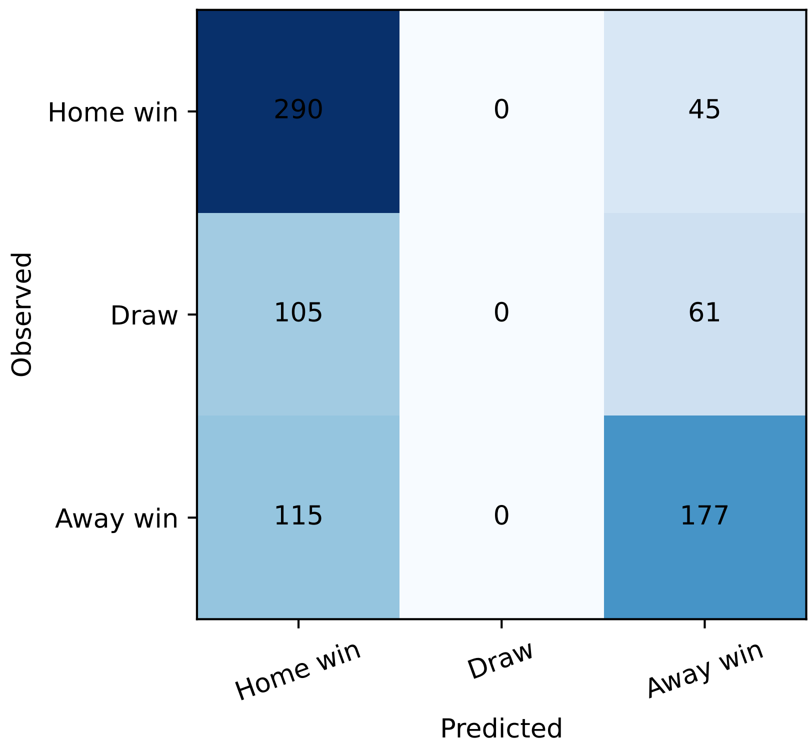

Figure 1: Held-out confusion matrix. The empty draw-prediction column explains the gap between accuracy and balanced metrics.

predictions. The model can still allocate nonzero draw probability, so proper scoring rules retain information not visible in the hard confusion matrix; nevertheless, a deployment requiring draw identification would be unacceptable.

The confidence plot in Figure 2 places most bins above the diagonal, indicating under-confidence after scaling. The two highest-confidence bins contain only 25 observations, so their apparent reliability is uncertain.

6.3 Feature analysis

Permutation importance on the holdout ranks away five-match mean goals, home five-match goal difference, home venue goal difference, and Elo difference as the most influential variables (Figure 3). Importances are small and correlated predictors can share or mask importance; the plot should not be interpreted causally.

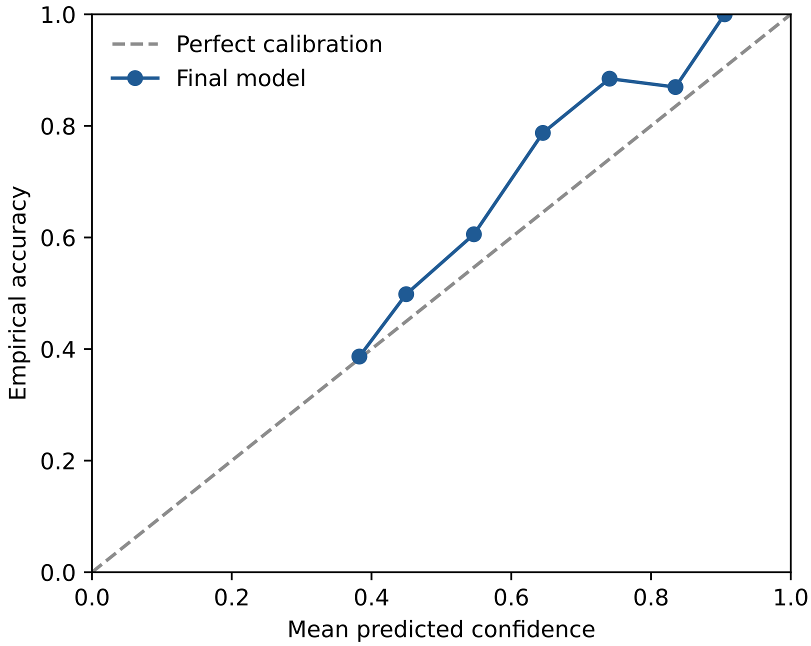

Figure 2: Confidence calibration on the chronological holdout. Point sizes are not weighted; the associated CSV records counts for each bin.

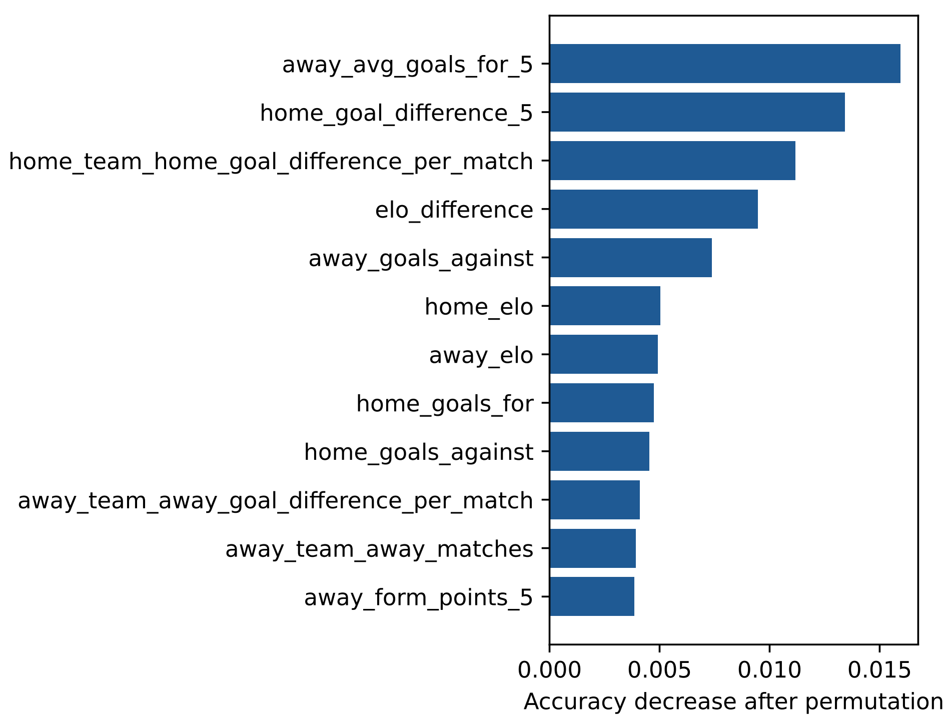

Figure 3: Top twelve permutation importances, measured as held-out accuracy decrease after 20 random permutations.

7 Limitations and Responsible Use

First, StatsBomb Open Data is curated, so the chronological holdout changes not only time but its mixture of competitions and seasons. Performance is not a claim about all soccer. Second, resetting state by competition-season discards continuity and transfers; cumulative counts can also encode season progress. Third, the model omits lineups, injuries, coaching changes, market information, and event-derived expected goals. Fourth, hard draw failure makes raw accuracy incomplete. Fifth, calibration bins are coarse and the fitted temperature does not transfer cleanly to the later period. Sixth, permutation analysis is associational.

No model was promoted or connected to decisions. The repository explicitly excludes real-money use. Any extension should pre-register a later temporal test, improve draw modeling (for example, ordinal or score-based formulations), quantify uncertainty across competitions, and test calibration by subgroup.

8 Reproducibility

The repository includes normalized input and derived Parquet tables; SHA-256 manifests; Python source; unit tests; all 793 held-out predictions; complete cross-validation and temperature sweeps; model and data cards; serialized fitted model; CSV tables; PDF/PNG figures; and the present LaTeX source. The experiment runs with:

```
python -m soccer_final.features
python -m soccer_final.experiment
```

The recorded run used Python 3.13.12, NumPy 2.5.1, pandas 3.0.5, and scikit-learn 1.9.0 on Apple Silicon macOS. Tests verify leakage ordering and metric utilities. Artifact hashes allow later readers to distinguish regenerated results from this submitted run.

9 Presentation Video Demonstration

To make the model workflow visible in the final presentation, we built a separate video demonstration over a 130.33-second, 1920×1080 FIFA highlight clip of Argentina versus Switzerland. Ultralytics YOLOv8m, pretrained on COCO, detects persons and sports balls on every second source frame at a 960-pixel inference size. A transparent HSV jersey-color heuristic labels likely Argentina and Switzerland players and leaves uncertain detections as generic persons. The ball was detected on 1,163 of 1,953 sampled frames (59.55%); 25,090 person and 1,865 ball detections were retained.

The overlay incorporates the submitted prematch model rather than presenting a disconnected computer-vision demo. Because the versioned training snapshot ends in 2025, the 2026 fixture has no valid competition-season history. The model is therefore evaluated at documented neutral new-season state, producing 46.59% Argentina, 20.28% draw, and 33.14% Switzerland. These fixed values appear under the larger display. The changing "current" visualization adds bounded, four-second-smoothed evidence from team-colored player confidence and nearest-player ball proximity in log-probability space. It is explicitly labeled as a research visualization rather than calibrated betting odds.

Inference ran locally on an Apple M1 Pro with 16 GB unified memory using the PyTorch MPS backend. The full pass took 146.37 seconds (13.34 sampled frames/s). A lossless H.264 4:4:4 annotated master was rendered before two-pass delivery compression. The final 1080p H.264/AAC artifacts are 178.76 MiB for the 200 MB cap and 44.19 MiB for the 50 MB cap. Both were decoded end to end and visually checked across wide play, set pieces, close-ups, and the end card. The source clip declares no downloadable license in YouTube metadata; it is excluded from Git, attributed on screen, and the derived presentation is documented for local course use subject to permission and fair-use requirements.

10 Conclusion

A carefully regularized linear model provides a strong, auditable baseline for prematch soccer probabilities: it substantially beats a class-prior baseline on the later chronological period and outperforms the more flexible candidates during time-ordered validation. The complete evaluation changes the interpretation, however. No hard draws are predicted, class-balanced metrics lag raw accuracy, and training-selected temperature scaling degrades later calibration. The main result is therefore twofold: leakage-safe form and strength features contain useful signal, but a credible final project must expose probability and class failures that a single accuracy number would conceal.

References

[1] Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

[2] Arpad E. Elo. *The Rating of Chessplayers, Past and Present*. Arco, New York, 1978.

[3] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[4] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330, 2017.

[5] StatsBomb. Statsbomb open data. `https://github.com/statsbomb/open-data`, 2026. Accessed 22 July 2026.